Connector for ODK - User Manual

Version 2.0

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Homepage: https://github.com/fnmutua/connector-for-ODK

1. Introduction

Connector for ODK is a QGIS plugin with four tools:

Tool	Purpose
Get Data	Download ODK Central form submissions and load them as map layers
Split Layer	Split a vector layer into separate layers by attribute value
QA/QC	Run quality checks on File Geodatabase layers and export issue reports
Import	Upload QGIS vector features to a KeSMIS server

Each dialog includes a collapsible Help panel. Click << Show Help to open it.

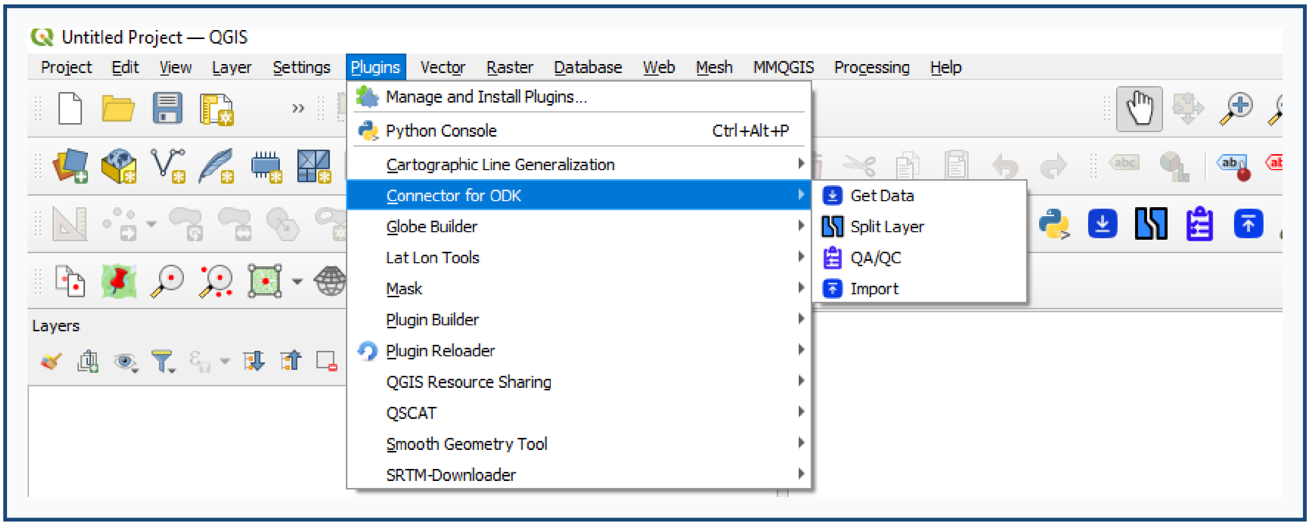

Figure 1 - QGIS menu and toolbar with Get Data, Split Layer, QA/QC, and Import

2. Prerequisites

2.1 QGIS

- QGIS 3.0 or later (desktop)
- An internet connection for ODK Central, KeSMIS, and optional package installation

2.2 Python packages

The plugin installs missing packages automatically the first time it loads. Required packages:

Package	Used for

numpy	Data processing
pandas	Tables and spreadsheets
geopandas	Spatial data handling
fiona	Reading/writing geospatial files
shapely	Geometry operations
fpdf	QA/QC PDF reports
pyproj	Coordinate reference systems

> Tip: QGIS ships with many geospatial libraries. If automatic installation fails, open OSGeo4W Shell (Windows) or your QGIS Python environment and run:

```
python -m pip install numpy pandas geopandas fiona shapely fpdf pyproj
```

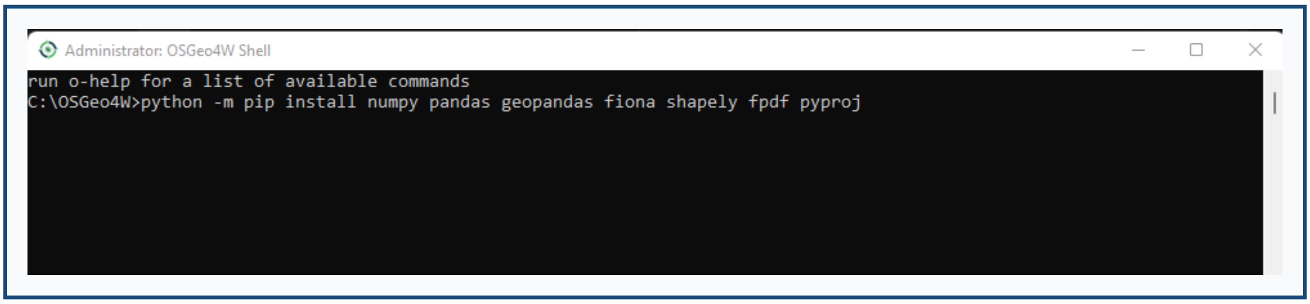

Figure 2 - Manual installation of Python packages (OSGeo4W Shell or QGIS Python console)

2.3 Data and access

Depending on which tools you use, you may also need:

- Get Data - ODK Central URL, username, and password
- Split Layer - Vector layers already loaded in the QGIS project
- QA/QC - An ESRI File Geodatabase (.gdb folder)
- Import - KeSMIS server URL, username, and password; vector layers loaded in QGIS

3. Installing the Plugin

Option A - QGIS Plugin Repository (recommended)

1. Open QGIS.
2. Go to Plugins -> Manage and Install Plugins.
3. Search for Connector for ODK.
4. Click Install Plugin.
5. Restart QGIS if prompted.

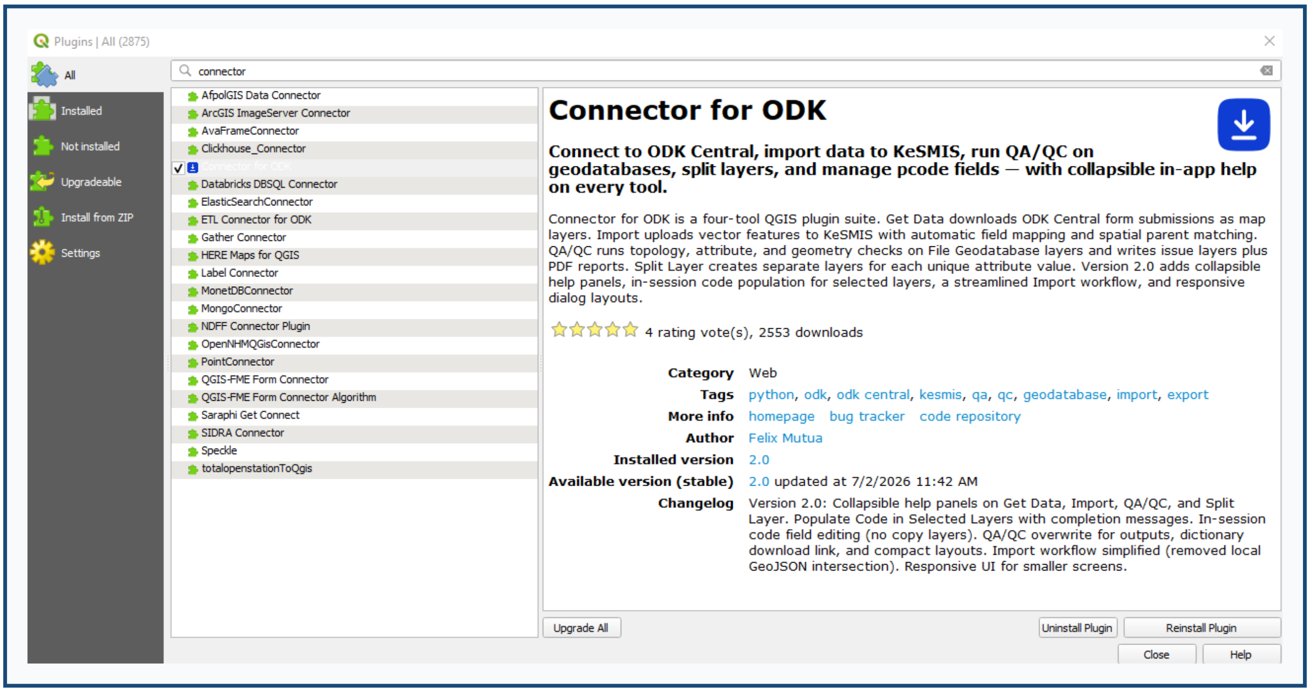

Figure 3 - Plugin Manager with Connector for ODK selected

Option B - Install from ZIP

1. Download connect_odk.zip (version 2.0).
2. In QGIS, go to Plugins -> Manage and Install Plugins.
3. Open the Install from ZIP tab.
4. Select the ZIP file and click Install Plugin.
5. Restart QGIS if prompted.

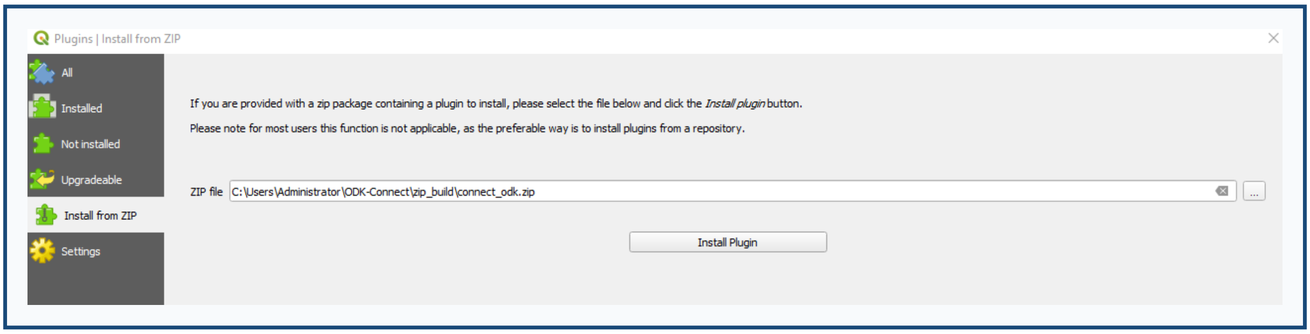

Figure 4 - Installing the plugin from a ZIP file

Verify installation

After QGIS restarts, you should see:

- Menu: Plugins -> Connector for ODK
- Toolbar icons for Get Data, Split Layer, QA/QC, and Import

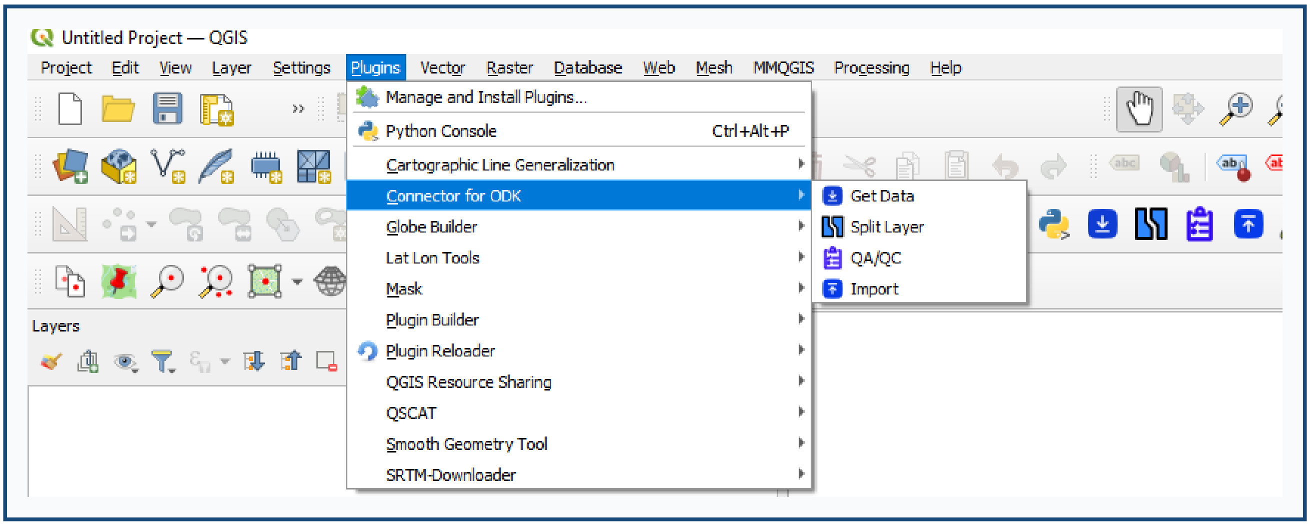

Figure 5 - Plugins menu with all four tools listed

4. Get Data (ODK Central)

Download ODK form submissions and add them to your map.

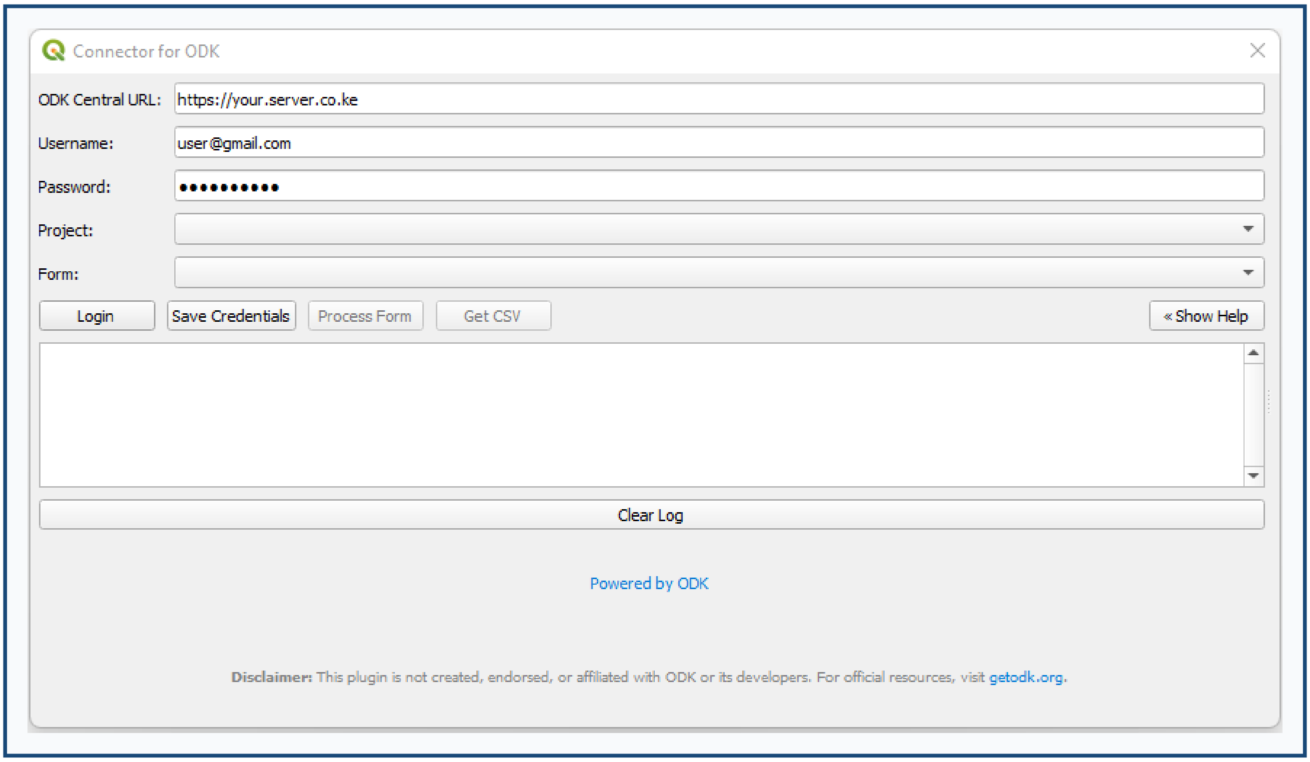

Figure 6 - Get Data dialog with ODK Central login

Steps

1. Open Plugins -> Connector for ODK -> Get Data.
2. Enter your ODK Central URL, username, and password.
3. Click Login to load projects and forms.

4. Select a project and form.

5. Click Process Form to fetch submissions and add a GeoJSON layer to the map.

6. Optionally click Get CSV to export the data as a spreadsheet.

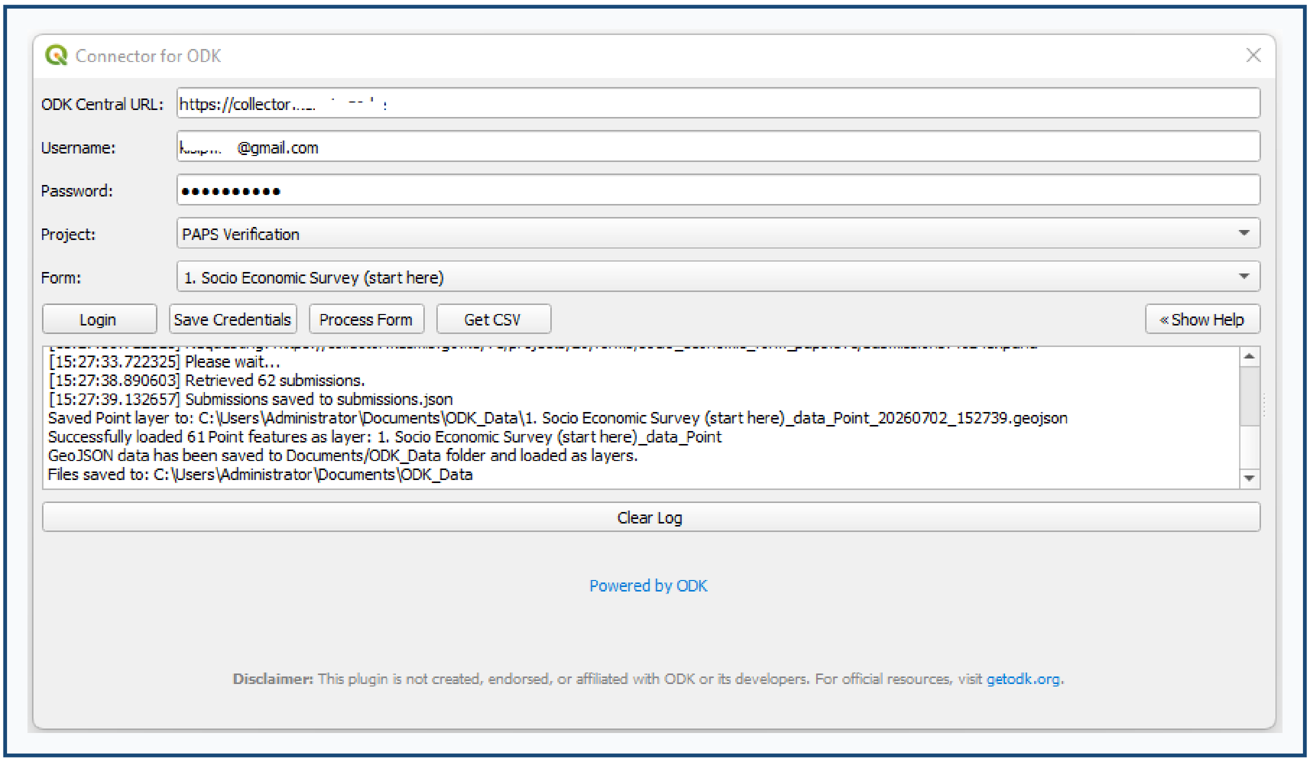

Figure 7 - Project and form selected before processing

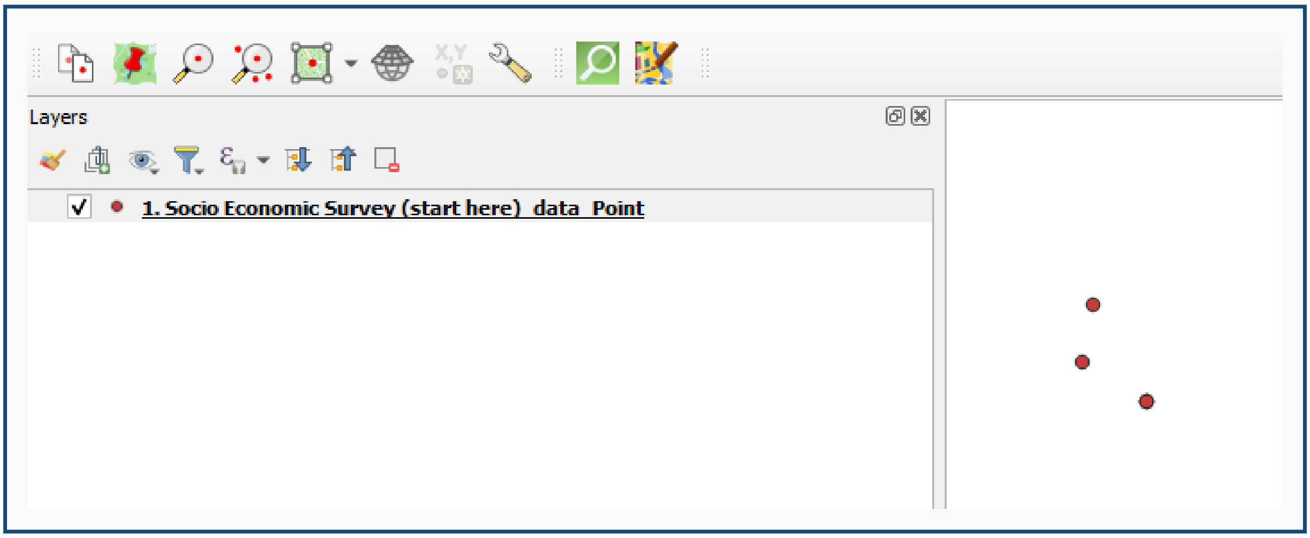

Figure 8 - Submissions loaded as a layer on the QGIS map

Notes

- Use Save Credentials to store your URL and login for next time.

- Output is in EPSG:4326 (WGS 84).

- A submissions.json file is written to the working folder.

- Check the Log panel at the bottom of the dialog for progress and errors.

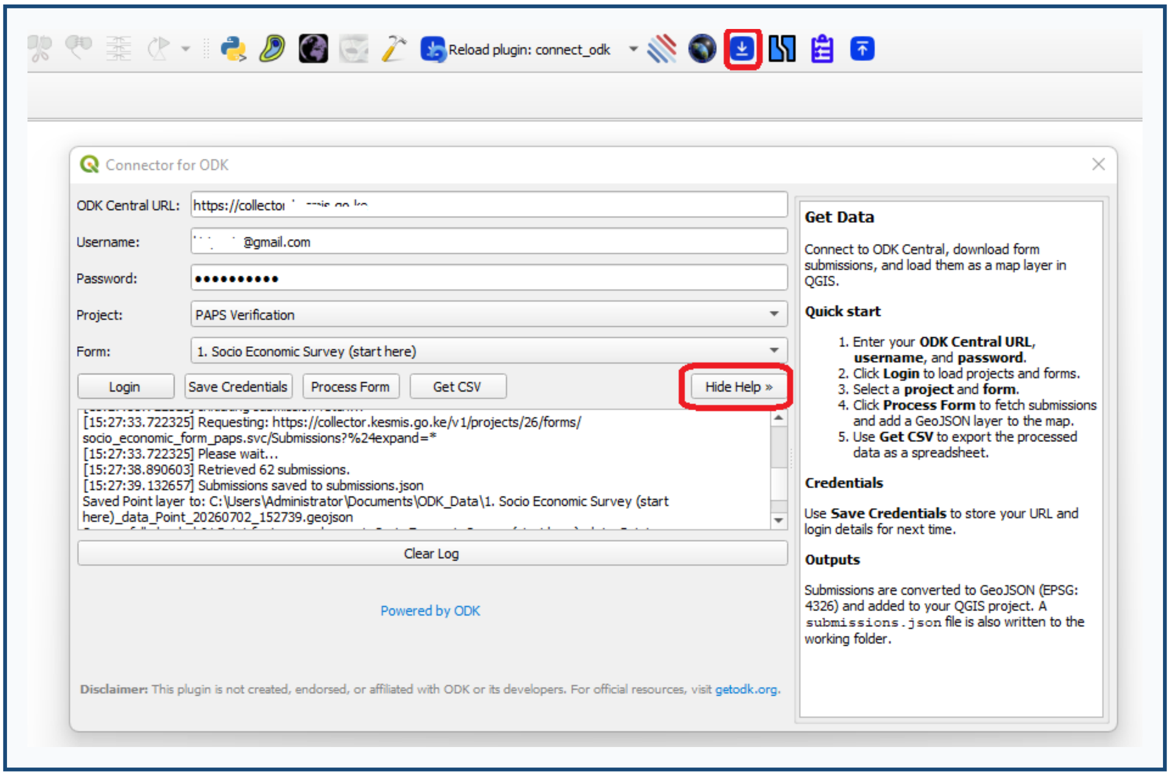

Figure 9 - Collapsible help panel in Get Data

5. Split Layer

Create separate in-memory layers for each unique value in a chosen attribute field.

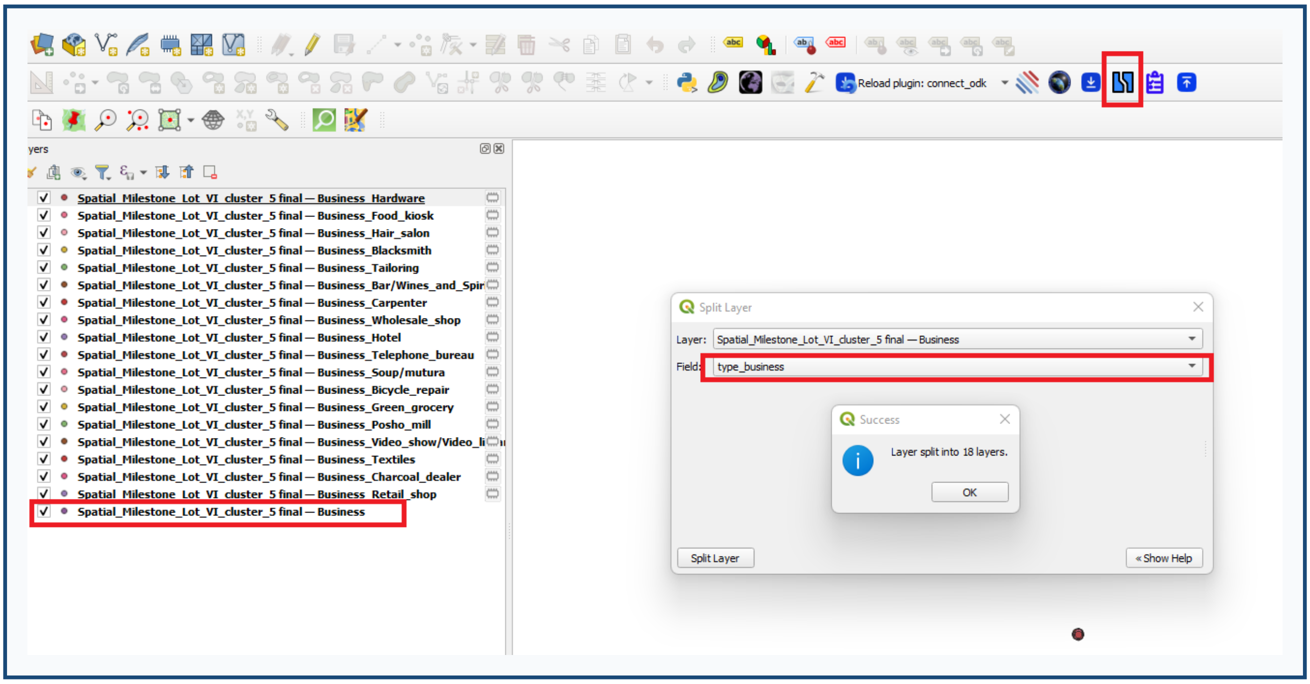

Figure 15 - Split Layer dialog

Steps

1. Load the source vector layer in QGIS.
2. Open Plugins -> Connector for ODK -> Split Layer.
3. Choose a Layer from the project.
4. Choose the Field to split on.
5. Click Split Layer.

Result

See Figure 15 for the Split Layer dialog. After splitting:

- One new layer is created per unique non-null value.
- Layers are named {layer}_{value}.
- Empty fields are dropped from each split layer.
- Geometry and CRS are copied from the source.

6. QA/QC

Run quality checks on File Geodatabase layers and export issue layers, spreadsheets, and a PDF summary.

Steps

1. Open Plugins -> Connector for ODK -> QA/QC.
2. Click Select GeoDatabase and choose the folder containing your .gdb.
3. Click Select Output Folder for reports and issue layers.
4. Adjust parameters if needed (angle and length thresholds).
5. Under Select Layers, tick the layers to check, or use Select All.
6. Click Run All Checks.
7. When finished, use the PDF Report and Open Output Folder links below the log.

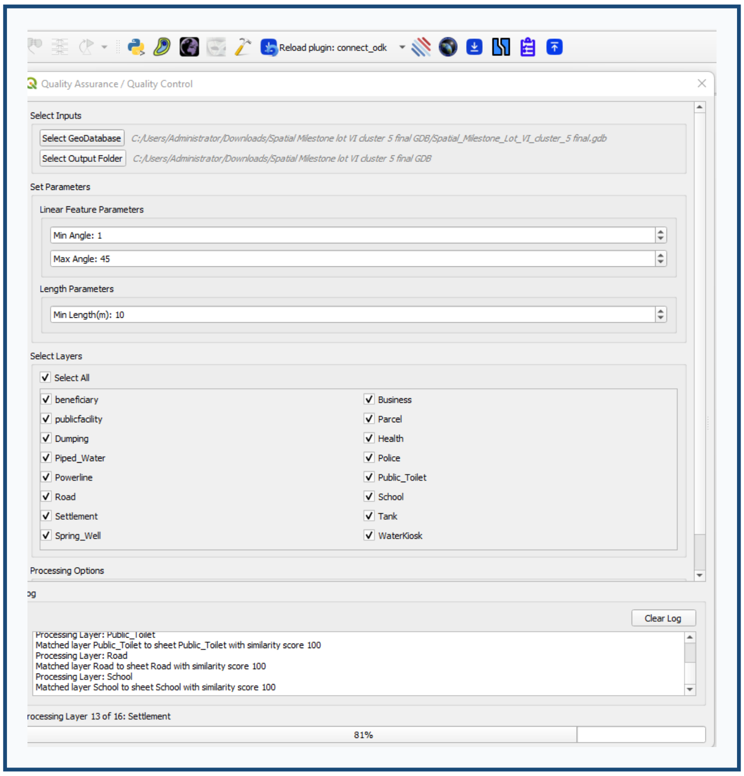

Figure 12a - QA/QC interface

Checks performed

Check	Description
Duplicate geometries	Features with identical geometry
Duplicate attributes	Rows with identical non-geometry fields
Overlapping polygons	Polygon pairs sharing area above 0.01 m2
Line issues	Sharp turns and self-intersections
Short lines	Line features shorter than the minimum length
Attribute issues	Fields validated against dictionary.xlsx

Parameters (defaults)

Parameter	Default	Purpose
Min Angle	1 deg	Lower bound for flagged turn angles
Max Angle	45 deg	Upper bound for flagged turn angles
Min Length	10 m	Flag lines shorter than this

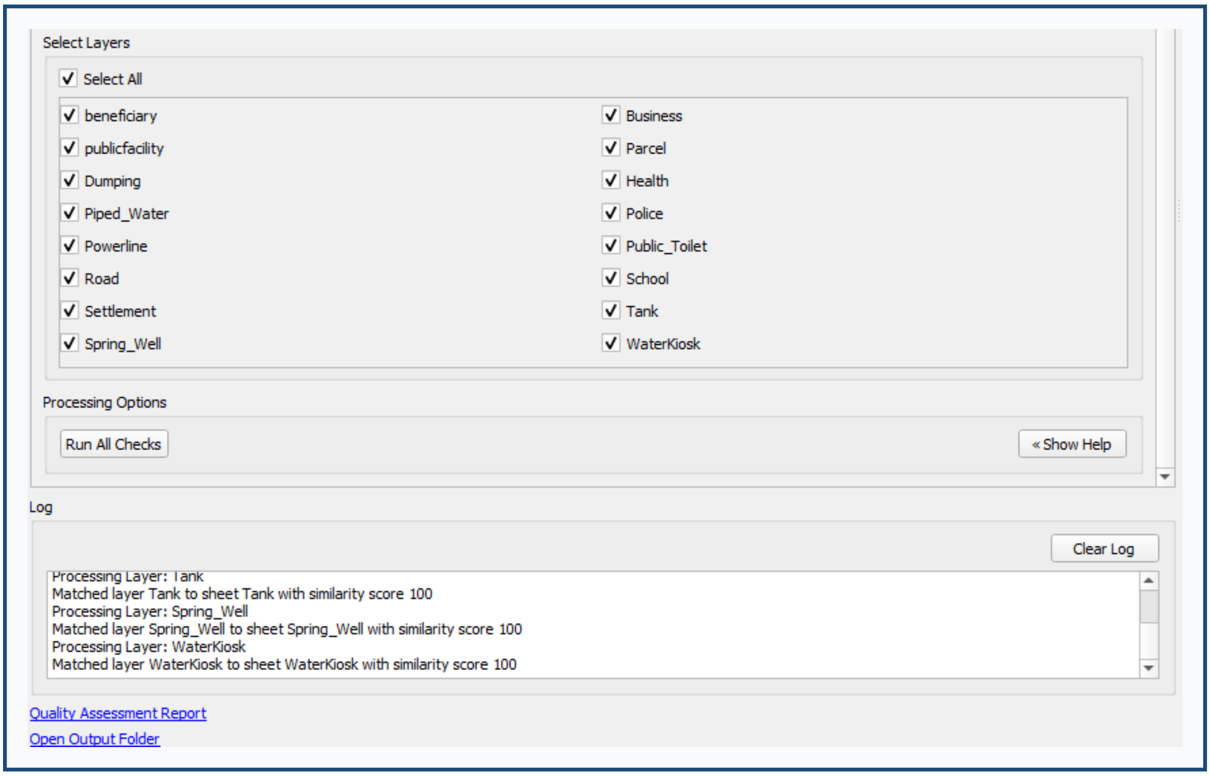

Figure 12b - Completed run with report and output folder links

Attribute dictionary

dictionary.xlsx is bundled with the plugin. Each sheet should match a layer name and include columns Attribute, Type, and optionally LEN and Options. You can download a copy from the help panel link inside the QA/QC dialog.

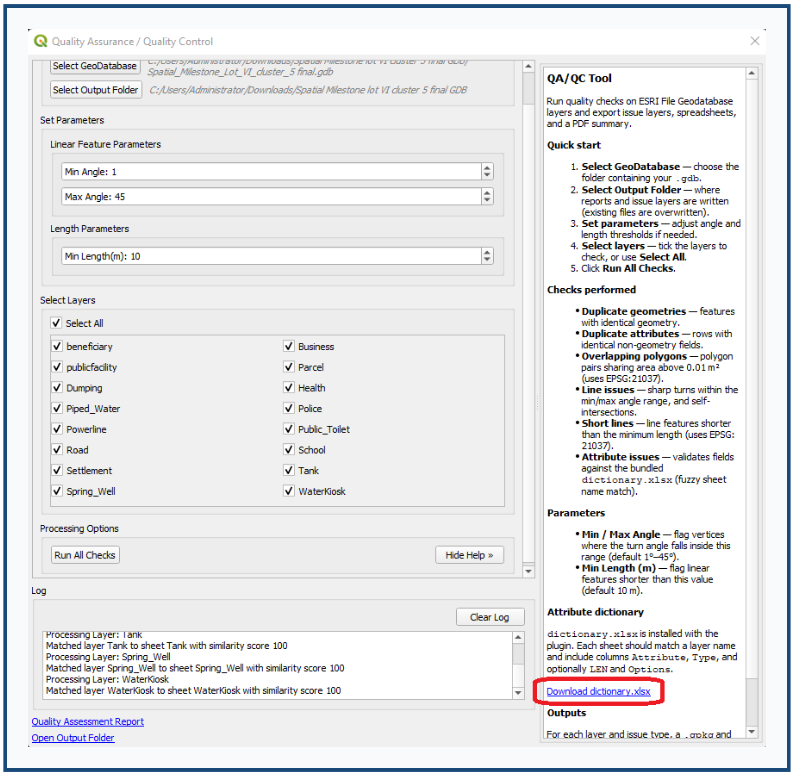

Figure 13 - Help panel showing dictionary download link

Outputs

For each layer and issue type, .gpkg and .xlsx files are written to the output folder. A summary PDF (database_summary_report.pdf) is also created. Existing output files are overwritten on re-run.

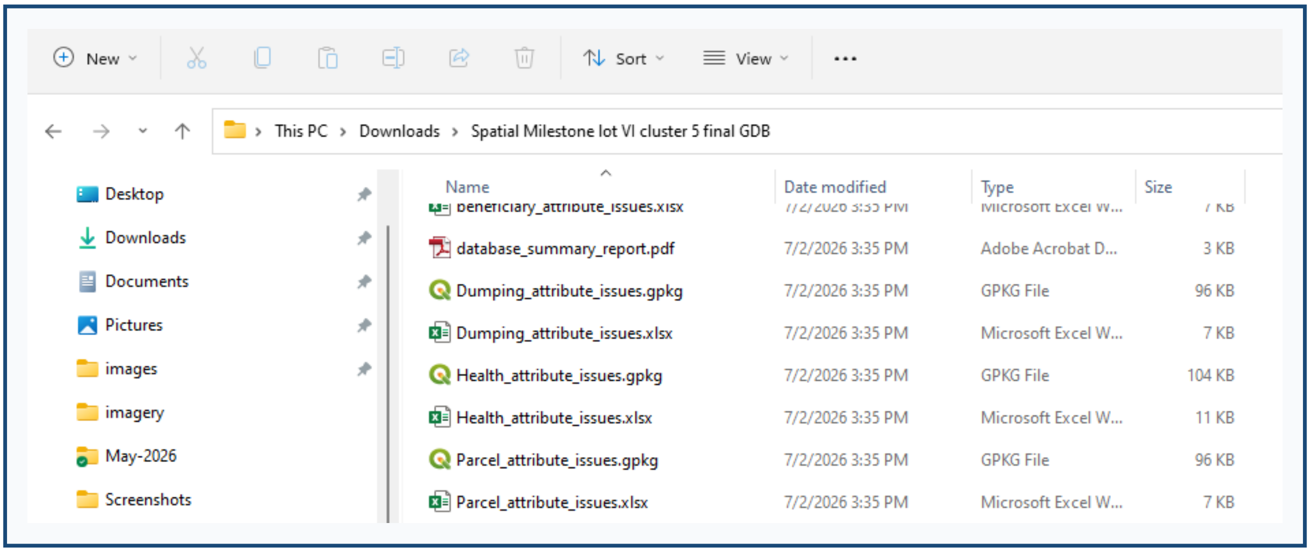

Figure 14 - Example QA/QC output files in the output folder

7. Import (KeSMIS)

Upload QGIS vector layer features to a KeSMIS server with automatic field mapping.

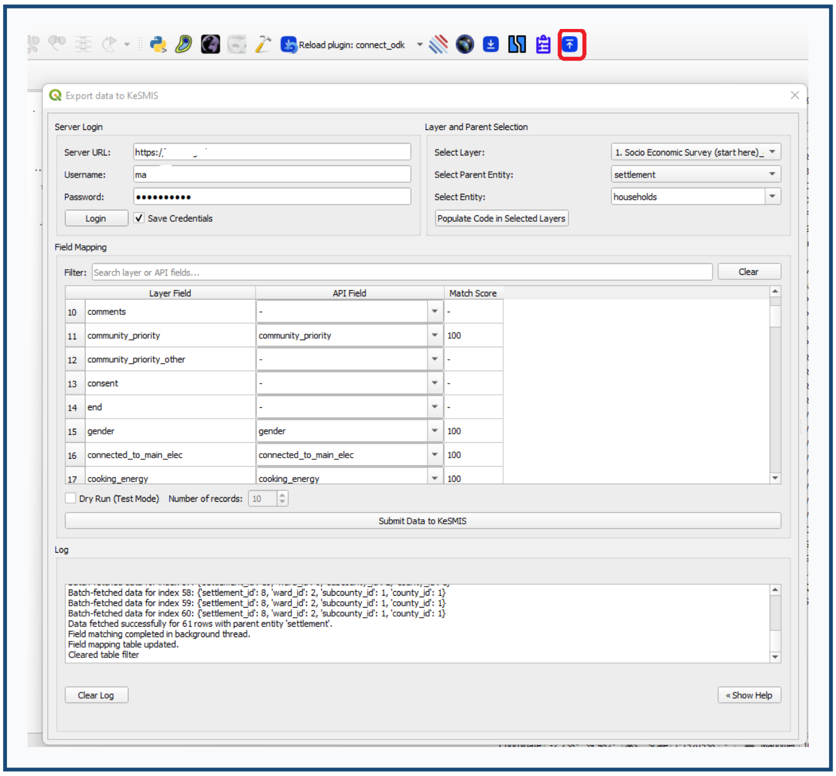

Figure 10 - Import dialog with layer, entity, and field mapping

Steps

1. Load your vector layer(s) in QGIS first.
2. Open Plugins -> Connector for ODK -> Import.
3. Server Login - enter the KeSMIS URL, username, and password, then click Login.
4. Select Layer - choose the QGIS vector layer to export.
5. Select Parent Entity - choose settlement or ward for spatial matching.
6. Select Entity - choose the API entity/model to submit to.
7. Review the Field Mapping table (see Figure 10). Use the filter box to search fields and adjust API field selections if needed.
8. Click Submit Data to KeSMIS.

Code column

Layers should include a code column for pcode matching.

1. Click Populate Code in Selected Layers.
2. Tick the layers to update and click Run.
3. A message will confirm success or report any skipped layers. Details appear in the log.

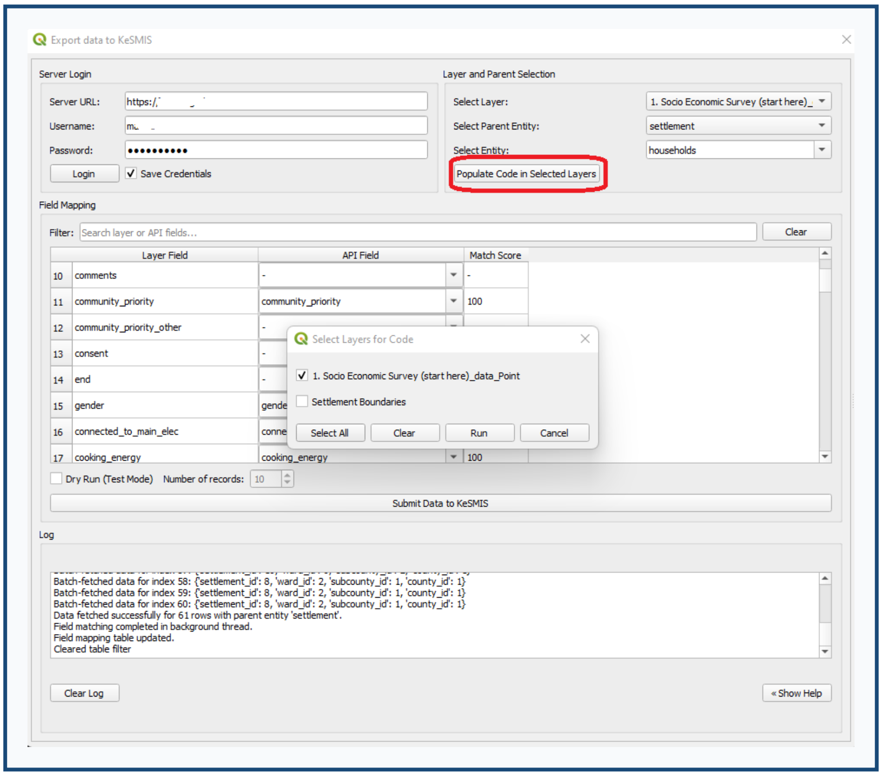

Figure 11 - Layer selection dialog for Populate Code

Dry run

Enable Dry Run (Test Mode) to submit only a limited number of records before a full upload. Set the record count in the spin box next to the checkbox.

8. Tips and Troubleshooting

Issue	What to try
Plugin does not appear after install	Restart QGIS. Check Plugins -> Manage and Install Plugins -> Installed that Connector for ODK is enabled.
Package install fails	Install dependencies manually using the QGIS Python environment (see Section 2.2).
ODK / KeSMIS login fails	Confirm URL, username, and password. Check network access to the server.
Import missing code field	Use Populate Code in Selected Layers before submitting.
QA/QC attribute check skipped	Ensure dictionary.xlsx has a sheet matching the layer name.
No layers in a dropdown	Load vector layers into the QGIS project first.

For updates, bug reports, and source code:

- Tracker: https://github.com/fnmutua/connector-for-ODK
- Repository: https://github.com/fnmutua/ODK-Connect